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METHODOLOGIES AND APPLICATION

Global migration strategy with moving colony for hierarchical distributed evolutionary algorithms

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Abstract In the established hierarchical distributed evolutionary algorithms (HDEAs), object of global migration is individual. To obtain better solutions of concrete problems, global migration strategy with moving colony is proposed in this paper. In global migration based on the proposed strategy, migration object is subpopulation which moves between groups. Such global migration can increase the efficiency of thereafter local migration. Moreover, realizing it even needs no communication because it can be executed by regrouping subpopulations. In our experiments, the basement of parallelism is an EA for the Traveling Salesman Problem. Outcomes of HDEAs based on proposed scheme which have different global migration topology are compared with those of traditional ones on nine benchmark instances. The results show that a HDEA based on the proposed strategy having the ring global topology performs better than traditional HDEAs for high difficulty instances. However, the advantage of that having the random global topology is not so significant because of conflicting migrations arisen from this topology.

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1 Introduction

Among widely known types of parallel evolutionary algorithms(PEAs), the distributed evolutionary algorithms(DEAs) are the most popular optimization procedures. One important reason for their popularity is that they can be readily implemented in distributed memory multiple instruction multiple data (MIMD) computers (Alba and Tomassini 2002). In DEAs, the parallel operator, migration, produces individual exchanges between subpopulations (Herrera et al. 1999).

Hierarchical distributed evolutionary algorithms (HDEAs) are DEAs whose nodes are other simple DEAs, which are connected with each other and called basic DEAs (Herrera et al. 1999). In this paper, one colony which consists of all subpopulations in one basic DEA is called a group. In HDEAs, there exist two types of migration, local migration and global one (Herrera et al. 1999; Cantu-Paz 1998). The former occurs between subpopulations belonging to the same group while the latter between groups.

HDEAs were ever introduced in Cantu-Paz (1998) and formally proposed in Herrera et al. (1999). However, from then on, how to improve the migration operator of HDEAs is rarely further discussed. Though HDEAs have been applied in quite a few fields (Gonzalez et al. 2006; Karakasis et al. 2007; Liakopoulos et al. 2008; Choi et al. 2009; Yuan and Bai 2010; Lee et al. 2011 are recent papers on this topic), the applications of HDEA are by far fewer than those of simple DEAs. The main causes are as below. Firstly, HDEAs is more difficult to realize than simple DEAs for their hierarchical structure. More importantly, it is hard to find a formal comparison on performance in literature to show the advantage of HDEAs. In this paper, such a comparison is carried out to show that our research is meaningful.

The main motivations of this paper are as follows. Our experiment to compare results of a DEA and those of a HDEA which are based on the same EA and have the same parameter settings shows the superiority of the latter. With the rapid development of hardware technology, platforms fitting the requirement of HDEAs are becoming more and more common. Against this background, improving solution quality of HDEAs for concrete problems is a meaningful task. In existing HDEAs, object of global migration is individual just as that of local migration. Such global migration can be called individual one. In fact, since the colonies between which global migration occurs is different in level form those between which local one does, migrating object in global migration can be different in level from that in local one, either.

Global migration strategy with moving colony is proposed for HDEAs in this paper. In this scheme, migrating object of global migration is not individual but subpopulation which moves between groups. In detail, once a subpopulation migrates to another group, its clone in the source group should be replaced by an immigrating subpopulation. After such global migration, in each group, immigrating subpopulations take part in thereafter local migrations. In experiments, outcomes of HDEAs based on this scheme different in global migration topologies are compared with those of traditional ones on nine benchmark instances of the Traveling Salesman Problem (TSP) from Reinelt (1996). All the HDEAs are based on the same EA and are as same as possible in parameter settings. The results show that the HDEA based on the proposed strategy having ring global topology can obtain significantly better solutions than traditional algorithms, while that having random global topology have not so significant an advantage on solutions.

This paper is set out as follows: Sect. 2 begins with a brief introduction of related researches. In Sect. 3, the global migration strategy with moving colony is presented and then employed in HDEAs. After that, experiments and the analysis are in Sect. 4. Finally, a conclusion and a prospect are dealt with in Sect. 5.

2 Related researches

2.1 Classification of distributed evolutionary algorithms

Parallel implementations of EAs were divided into four categories, master-slave PEAs, coarse-grained ones (DEAs), fine-grained ones and hierarchical ones (Alba and Tomassini 2002; Cantu-Paz 1998). All two level approaches of parallelization each uses a combination of the first three categories were generalized into the fourth one. Based on the above work, DEAs can be classified into two types. One is simple DEAs, and the other is HDEAs. A HDEA is a two level combination of DEAs. The simple DEA is depicted in Fig. 1. Meanwhile, Fig. 2 shows the HDEA.

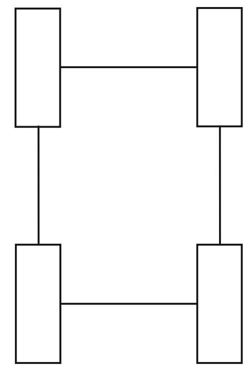

Fig. 1 Sketch of simple DEA. Subpopulations send/receive individuals infrequently to/from others

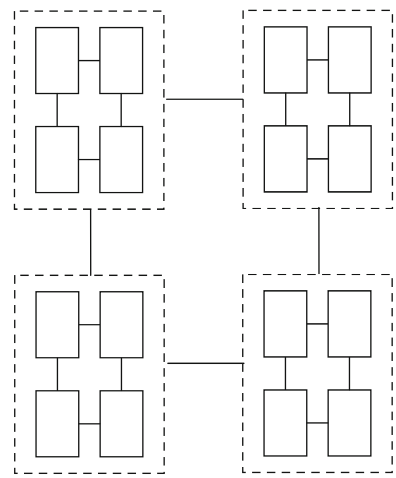

Fig. 2 Sketch of HDEA. In local migration, subpopulations send/receive individuals to/from others in the same group. More sparsely, subpopulations send/receive individuals to/from other groups in global migration

2.2 Distributed evolutionary algorithms

DEAs are based on the idea of a single population divided into multiple subpopulations (Power et al. 2005). Each subpopulation evolves independently and simultaneously. When the given criterion is met, migration exchanges individuals between subpopulations (Herrera et al. 1999; Cantu-Paz 1998). In the majority of DEAs, migration occurs after predetermined constant intervals (Cantu-Paz 1998). Each immigrant to a subpopulation will replace an individual in it. On one hand, the semi-isolation of subpopulations can maintain difference among them. On the other hand, migration affords new building blocks for subpopulations at intervals (Cohoon et al. 1987). In conclusion, independent evolution and migration at intervals working together delays the stagnation of evolution.

Using such a coarse-grained parallelization can have advantages as below (Lässig and Sudholt 2013):

- Compared to parallelizing function evaluations, parallelization in this approach requires very little overhead because the amount of communication between different processes is very low.
- The effort of managing a small population can be much lower than the effort of managing a large, panmictic population.

Main parallel parameters for DEAs are listed as below:

- Migration topology, it describes which subpopulations send individuals to which ones (Skolicki and De Jong 2005). It is always a focus in the research of DEAs. For instance, Arnaldo et al. (2013) is a recent paper which performed a systematic analysis of the correlation between migration topology and problem structure.
- Migration strategy, it defines how to select emigrants and individuals to be replaced in each subpopulation. All existing migration strategies can be divided into two types (Herrera et al. 1999). Some belong to copying methods (Herrera et al. 1999) in which the clone of each emigrant in the source subpopulation is not specially treated. Best-worst and random-random are two common ones of them (Skolicki and De Jong 2005). The others belong to moving ones (Herrera et al. 1999) in which the clone must be replaced by an immigrant.
- Migration size, it specifies the quantity of individuals emigrating from or immigrating to a subpopulation during one migration round.
- Migration interval, it is the every certain number of generations between a migration round and the next one (Skolicki and De Jong 2005).
- Subpopulation size, it is the quantity of individuals in each subpopulation.
- Subpopulation quantity, it is the number of subpopulations in the population.

To interpret a phenomenon in DEAs, some new concepts are defined as below:

Definition 1 Similarity degree of S_θ (subpopulation θ) and S_μ, in the narrow sense, it means that the quantity of individuals which belong to S_θ and have clones in S_μ as a percentage of the subpopulation size.

Definition 2 Invalid migration, in the narrow sense, it means that the target subpopulation of migration has some individuals which are as same as the coming ones.

Invalid migration cannot afford any new building blocks but may make redundancy. In fact, the possibility of invalid migration is proportional to the similarity degree of the source subpopulation and the target one. Inevitably, the similarity degree gradually increases during a run because migration exchanges individuals among subpopulations at intervals. As a result, invalid migration appears more and more frequently.

2.3 Hierarchical distributed evolutionary algorithms

The existence of the two different types of migration is the key feature of HDEAs since they establish the real hierarchy between basic DEAs and the HDEA (Herrera et al. 1999). Individuals migrate between subpopulations in local migration, while ones migrate between groups in individual global migration. In individual global migration, each immigrant from its source subpopulation in the source group replaces an individual of the target subpopulation in the target group. In practice, local migration occurs much more frequently than global one (Herrera et al. 1999; Cantu-Paz 1998). Under such a scenario, there is a strong possibility that two subpopulations belonging to different groups have significantly lower similarity degree than two ones in the same group. Consequently, it is highly possible that an immigrant of global migration has new building blocks for the target group. Then, it and its offspring may be useful to decrease the similarity degree of subpopulations in this group and then help to enhance efficiency of following local migration. According to the above analysis, the hierarchy leads to better convergence. However, there are few experiments comparing solving ability of HDEAs and that of DEAs directly in literature till now. Hence, such an experiment to prove the above analysis is carried out in Sect. 4.3 of this paper.

Ratio of local migration rounds to a global migration round, which decides the total times of both local migration and global one during a run together with the migration interval and the migration size, is a specific parameter of HDEAs. Moreover, the migration size, topology and strategy of HDEAs all include both a global aspect and a local one. In fact, existing global migration strategies are only different emigration-replacement methods for individual global migration. In other words, they are plans about how to select the emigrating individual and the replaced one for global migration. The subpopulation quantity of a HDEA can be expressed as $M \times N$. M is the quantity of groups. N is the quantity of subpopulations in each group.

3 Global migration strategy with moving colony

3.1 Definitions

In HDEAs, subpopulations consisting of individuals are relatively independent colonies for local migration. Object of local migration is individual. Likewise, groups consisting of subpopulations are relatively independent colonies for global migration. Object of traditional global migration is individual, too. It can be seen that migration object is two grades lower than colonies between which migration occurs in individual global migration, while one grade lower in local one. If the grade difference in global migration is set as same as that in local migration, object of global migration should be subpopulation.

Hence, not only an emigration-replacement method, but also the grain size of migration object should be taken into consideration in a global migration strategy. Then, colony global migration strategy, which regards subpopulation as global migration object, is defined as follow:

Definition 3 Colony global migration strategy, it is a type of global migration strategy which specifies that migration object is not individual but subpopulation.

Figure 3 is the sketch of individual global migration, while Fig. 4 is that of global migration based on the colony global migration strategy.

Nevertheless, Definition 3 only gives the grain size of migration object. An emigration-replacement method is required to complete a global migration strategy. In practice, copying methods have been used in the overwhelming majority of established DEAs and HDEAs. However, colony global migration based on a copying method may make identical subpopulations appear in different groups. It runs counter to the fundamental principle that difference should be maintained among colonies. On the contrary, no copy will happen in colony global migration based on a moving method. Therefore, moving methods are much better choices than copying ones for the colony global migration strategy. Colony global migration strategy based on a moving emigration-replacement method or, in short, global migration strategy with moving colony is defined as follow:

Definition 4 Global migration strategy with moving colony, it means that, in a global migration round, each subpopulation immigrating to a group replaces the clone of an emigrating subpopulation in this group.

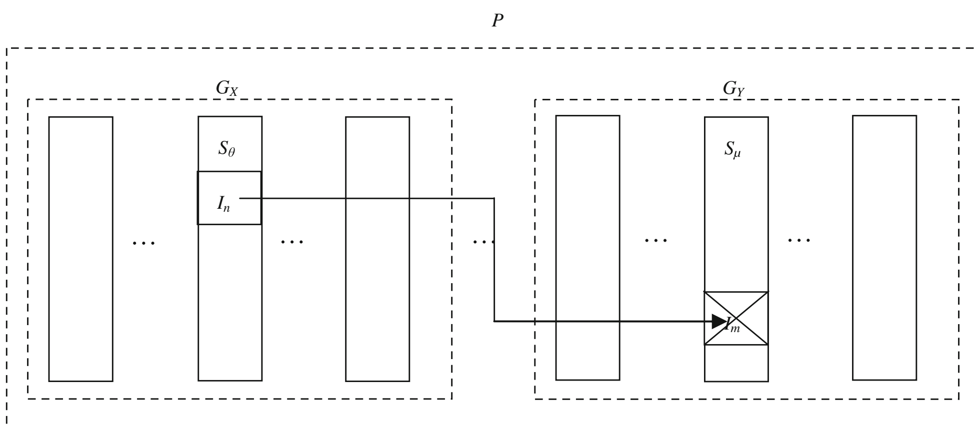

Fig. 3 Sketch of individual global migration. P is the population. G_X denotes group X. S_θ denotes subpopulation θ. I_n denotes individual n

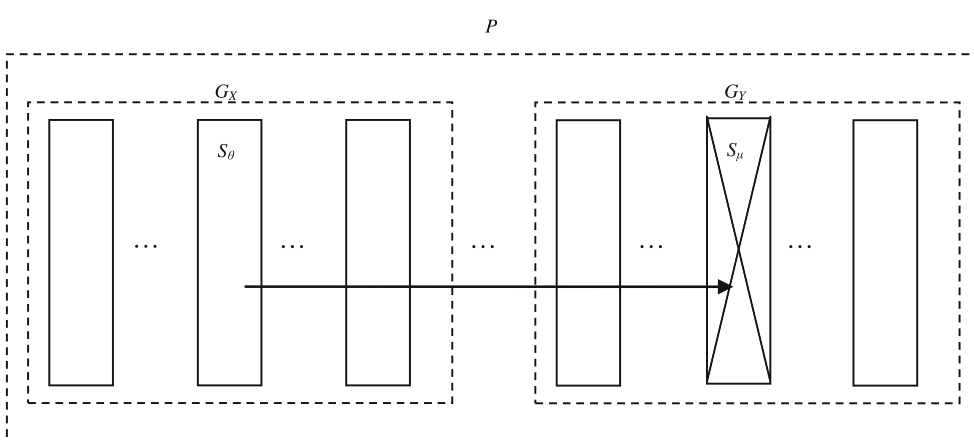

Fig. 4 Sketch of colony global migration. P is the population. G_X denotes group X. S_θ denotes subpopulation θ

3.2 Different global migration based on proposed strategy

Global migration based on the proposed strategy can be called global migration with moving colony. Here are two examples as below.

Example 1 Let $M \times N$ subpopulations are included in P ($S_{X,\theta}$ denotes the No. θ subpopulation in G_X). In the first global migration round, $S_{1,1}$ migrates to G_2, $S_{2,1}$ migrates to $G_3 \cdots S_{M,1}$ migrates to G_1. At the same time, in each group, the clone of emigrant is replaced by the immigrant. In the second round, $S_{1,2}$ migrates to G_2, $S_{2,2}$ migrates to $G_3 \cdots S_{M,2}$ migrates to G_1. At the same time, clones are replaced by immigrants.... Every round is according to above rule. After a round in which $\{S_{1,N}, S_{2,N} \cdots S_{M,N}\}$ migrate, $\{S_{1,1}, S_{2,1} \cdots S_{M,1}\}$ will migrate in the next round.

Global migration as Example 1 can be called global migration with moving colony based on the "subpopulation in turn-emigrant" method and the ring topology or, in short, ring global migration with moving colony. According to Example 1, the global migration size is one. In rounds of such migration, each subpopulation of a group moves in sequence to the corresponding position in the next group. Its pseudo code is in Fig. 5.

Example 2 Let $M \times N$ subpopulations are included in P. Integer $\theta \in [1, M \times N]$ and $\mu \in [1, M \times N]$ are generated randomly with the condition that S_θ and S_μ are not in the same group. Then, S_θ and S_μ move to the position of each other. Above operation is executed multiple times in a migration round.

Global migration as Example 2 can be called global migration with moving colony based on the "random subpopulation-emigrant" method and the random topology or, in short, random global migration with moving colony. Its pseudo code is in Fig. 6.

```
1=>j
while the evolutionary condition is satisfied do
    ......
    if global migration condition is satisfied then
        1=>i
        while i≤M do
            S_{i,j} migrates to G_{i%M+1} and then replaces the clone of S_{i%M+1,j}
            i+1=>i
        end while
        j%N+1=>j
    end if
end while
```

Fig. 5 Pseudo code of ring global migration with moving colony

```
while the evolutionary condition is satisfied do
    ......
    if global migration condition is satisfied then
        1=>i
        while i≤specified_value do
            get random integer θ∈[1, M×N]
            do
                get random integer μ∈[1, M×N]
            while S_θ and S_μ are in the same group
            exchange the position of S_θ and S_μ
            i+1=>i
        end while
    end if
end while
```

Fig. 6 Pseudo code of random global migration with moving colony

3.3 Functional analysis

In global migration with moving colony, provided that settings are reasonable, an immigrating subpopulation should have a great segregation to local ones so as to have a relatively low similarity degree to them. As this subpopulation takes part in thereafter local migration, its individuals, which have a high probability of having new building blocks, immigrate to other subpopulations of this group in batches. Consequently, the efficiency of thereafter local migration can be promoted.

3.4 Comparison of communication cost

Let the size of an individual be b, the maximal generations be g, the migration interval be m, the ratio of local migration rounds to a global migration round be $n : 1$, the migration size be r and the quantity of groups be M. During a run of a traditional HDEA, the amount of communication data for global migration, V, can be measured by the formula below:

$$V = \left[\frac{g}{m \times (n+1)}\right] \times M \times r \times b \tag{1}$$

In a run of HDEAs, each subpopulation must be in only one computing (slave) process. In contrast, group is just a concept in logic. In our program of HDEAs based on the proposed scheme, a 2D array in the master process records each group

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Fig. 7 Initial state of array. *Each number* denotes a subpopulation. *Each row of number* expresses that a group contains which subpopulations

57	58	59	60	61	6	7	8
1	2	3	4	5	14	15	16
9	10	11	12	13	22	23	24
17	18	19	20	21	30	31	32
25	26	27	28	29	38	39	40
33	34	35	36	37	46	47	48
41	42	43	44	45	54	55	56
49	50	51	52	53	62	63	64

Fig. 8 State of array after rounds of ring global migration with moving colony. *Each number* denotes a subpopulation. *Each row of number* expresses that a group contains which subpopulations

having which subpopulations. Here are examples of the 2D array in a 8 $\times$ 8 HDEA as below. The initial state of this array is shown in Fig. 7. Then, Figs. 8 and 9 are the state after rounds of global migration with moving colony based on the different topologies, respectively. According to these figures, local migration only occurs between two subpopulations in the same row. If some numbers in the array move between rows, it means that there are subpopulations move between groups. By this way, global migration with moving colony can be executed without communication between the master process and slave processes. Hence, for a HDEA based on global migration with moving colony, the amount of communication data for global migration during a run, V', can be measured by the formula below:

$$V' = 0 \tag{2}$$

expresses that a group contains which subpopulations

27	35	3	4	5	52	7	8
9	10	11	12	24	29	15	16
17	18	19	20	21	22	23	13
25	59	1	28	14	30	31	32
44	34	2	51	37	38	39	40
41	42	43	33	45	57	47	48
49	50	36	62	53	54	55	56
46	58	26	60	61	6	63	64

Fig. 9 State of array after rounds of random global migration with moving colony. *Each number* denotes a subpopulation. *Each row of number* expresses that a group contains which subpopulations

4 Experimental studies

All experiments are carried out on a Drawing TC5000A Computing Platform, which has 1,264 2.6 GHz cores and 1.5 TB memory. According to the subpopulation quantity of a run, the same number cores are used. The algorithms in our experiments include:

- The ring topology DEA (ring DEA in short),
- The HDEA based on ring topology local migration and ring topology individual global migration (ring individual HDEA in short),
- The HDEA based on ring topology local migration and random topology individual global migration (random individual HDEA in short),
- The HDEA based on ring topology local migration and ring global migration with moving colony (ring colony HDEA in short),
- The HDEA based on ring topology local migration and random global migration with moving colony (random colony HDEA in short).

4.1 Problem set and simple EA

The TSP is a famous NP-hard problem. Reinelt (1996) has its benchmark instances. Most of them are difficult for EAs. However, some other algorithms, such as the Lin–Kernighan (LK) algorithm (Lin and Kernighan 1973), can obtain the optimal solution of most TSP instances. Hence, the optimal solution of most instances in Reinelt (1996) is given besides their data set. It should be stressed that the purpose of our experiments is not to obtain better solutions of any problem than ever but to test our scheme by comparing the performance of different HDEAs. Overall, the TSP is a fitting problem for our experiments.

An EA for the TSP adapted from the improved GT algorithm (Cai et al. 2005) is the basement of parallelism in our experiments. It is based on the inver-over operator (Tao and Michalewicz 1998) which has adaptive components serving as mutation or crossover. In detail, sometimes inversions applied to a single individual, while other times the segment of an individual to be inverted is determined by another randomly selected one. Besides the inver-over operator, another one, so called mapping in Cai et al. (2005), which belongs to crossover, is employed in the EA. The flow of mapping operator is as follow. Firstly, two individuals are selected at random. Then, a segment of chromosome in the individual having worse fitness in them is selected at random. After that, a segment which contains the same number of cities and has the same first city is searched in the other individual. This found segment replaces the former segment in the worse individual's chromosome. Finally, the other part of the chromosome is adjusted according to the last steps of partially mapped crossover (Goldberg and Lingle 1985) which is a traditional operator used in EAs for the TSP. In addition, critical velocity (Cai et al. 2005), which is a threshold value of the improving degree from the last best solution to the current one per unit time or, in short, improving velocity, is used in the EA to switch two evolutionary strategies. Mapping is executed at the given probability only if the improving velocity is below the critical value. In total, the differences between our EA and the improved GT algorithm in Cai et al. (2005) are listed as below. Firstly, the so-called optimization operator (Cai et al. 2005) is abandoned for its negative factor to the synchronization of processes. Also, the difference between the two evolutionary strategies becomes smaller to decrease the sensitivity of the critical velocity to different instances for outcomes. The pseudo code of the EA employed in our experiment is given in Fig. 10.

It is reported in Cai et al. (2005) that the improved GT algorithm can obtain the optimal solution of quite a few TSP instances smaller than a280. Hence, nine bigger instances in Reinelt (1996) from lin318 to pr1002 are selected for our experiments.

4.2 Experimental settings

To simplify the setup, setting of the critical velocity shifted from different values for different instances in Cai et al. (2005) to the same value for all instances in our experiment. Other evolutionary parameters are all set as same as Cai et al. (2005). Meanwhile, except the migration interval and the subpopulation quantity, all parallel parameters are fixed. Skolicki and De Jong (2005) and Fernandez et al. (2001) are references for parallel parameters settings. Five equal difference migration interval values are set for each instance. In the experiment to compare results of the ring DEA and the ring colony HDEA, the subpopulation quantity is set three different values for both algorithms.

The settings of parameters except the migration interval are listed in Table 1. It can be seen that both the migration topology and the emigration-replacement method of the ring HDEAs are certain, while those of the random HDEAs are stochastic. Moreover, it is notable in Table 1 that the global migration size of the random colony HDEA is different from that of the other algorithms. The reasons are listed as follows. Random global migration with moving colony has a very narrow range of choices on migration object. In our settings, the subpopulation quantity is 8×8. It means that each random choice must be made from eight subpopulations in the same group. Thus, conflicting migration happens at a high percentage. For instance, after the exchange of $S_{X,\theta}$ and $S_{W,\gamma}$, these two subpopulations soon exchange again. If the global migration size is too small, valid migration is not enough in every round. Therefore, the global migration size of the random colony algorithm is set slightly larger than that of other three algorithms to compensate for the waste of exchange chances by providing more ones.

$$p = p \times \left(1 - \frac{g_n}{g} \times 0.01\right) \text{ where } g_n \text{ denotes current generations and } g \text{ represents the maximal ones} \quad (3)$$

4.3 Experiment to compare results of DEA and HDEA

In this experiment, the ring DEA and the ring individual HDEA, both run thirty times independently for each instance at three subpopulation quantities under five migration intervals, respectively. The total running times are $2 \times 9 \times 5 \times 3 \times 30 = 8{,}100$ ones. The results under different subpopulation quantities are shown in Table 2.

In Table 2, averages show that solutions of both algorithms become better and better with the raise of the subpopulation quantity. According to the t-test result in the table, the ring colony HDEA significantly wins the ring DEA in 17 cases when the subpopulation quantity is 16. When 36 subpopulations are employed, the ring colony HDEA significantly wins in 18 cases. After subpopulation quantity goes to 64, the number is 20. Except the above cases under different subpopulation quantity, there is no significant difference in all other ones. It can be concluded that the ring individual HDEA do have significant better solving ability than the ring DEA. Further, the advantage of the HDEA goes more and more significant with the increase of the subpopulation quantity. In summary, the hierarchy does make migration more effective and then leads to better solutions.

Thus, if a DEA which has many subpopulations cannot get satisfactory solutions, better ones can be obtained by using

```
while the evolutionary condition is satisfied do
    for each individual in the population do
        product a copy of the current individual, x
        randomly select a city, c, in the chromosome of x
        do
            if a random pure decimal is less than a parameter, p, then
                randomly select another city, c', in the chromosome of x
            end if
            else in the chromosome of another randomly chosen individual in the population except the current one, let the city next to c
          be c'
            if c and c' are neighbours in x
                break out of the do-while loop
            end if
            inverse the segment of the chromosome of x from the city next to c to c'
            if the fitness of x is better than the current individual
                the current individual is replaced by x
                break out of the do-while loop
            end if
            let c' be c
        while there still exist cities not be regarded as c
    end for
    if the current improving velocity is less than the critical value and a random pure decimal is less than a parameter, p', then
        two individuals are selected at random
        a segment of chromosome in the individual having worse fitness in them is selected at random
        a segment which contains the same number of cities and hasthe same first city is searched in the other individual
        this found segment replaces the former segment in the worse individual's chromosome
        the other part of the chromosome is adjusted according to the last steps of partially mapped crossover
    end if
end while
```

Fig. 10 Pseudo code of our EA

a HDEA based on the same EA having the same quantity of subpopulations. Running on the same platform, such an upgrade in algorithm even does not require extra running time because no more volume of computation or communication is demanded.

4.4 Connection mode experiment

Different from individual global migration, there is no individual entering or leaving any subpopulation when global migration with moving colony occurs. As a result, besides the common parameters, a new parameter, connection modes for a round of global migration with moving colony and the following local migration round or, in short, connection mode is required to be setup in the colony HDEAs. A choice should be made between two modes. One is named compact one in which the following local migration round is executed immediately after global migration with moving colony. The other is loose mode in which the following local migration round occurs one migration interval later. In the former mode, there are individuals entering and leaving subpopulations after every migration interval. Such a phenomenon does occur in traditional HDEAs. In the latter one, there is a migration interval between a global migration round and the following local migration round. Such a characteristic does exist in individual HDEAs. To make the right choice, comparisons are carried out between outcomes of the colony HDEAs based on different connection modes. In detail, out-

Table 1 Setting of parameters

Evolutionary parameters		
Initial mutation rate (P)	0.02 (changing during a run according to formula (3))	
Crossover rate	$1 - P$	
Mapping rate (Cai et al. 2005)	0.05	
Critical velocity	5,000	
Parallel parameters		
Subpopulation size	100	
Subpopulation quantity	64 (except for experiment in Sect. 4.3)	
Ratio of local migration rounds to a global migration round (HDEAs only)	20:1	
Local migration size	1	
Global migration size (HDEAs only)	Ring colony algorithm	1 (subpopulation)
	Ring individual algorithm	1 (individual)
	Random colony algorithm	2 (subpopulations) on average
	Random individual algorithm	1 (individual) on average
Emigration-replacement method of local migration	Random-worst	
Emigration-replacement method of global migration (HDEAs only)	Ring colony algorithm	According to Example 1
	Ring individual algorithm	The best individual of the subpopulation in turn-the worst one of the same subpopulation
	Random colony algorithm	According to Example 2
	Random individual algorithm	A random individual of a random subpopulation-a random individual of the same subpopulation
Terminal criterion	Running same generations as HDEA (for DEA) 100 global migration rounds having been done (for HDEA)	

comes of the ring colony HDEA based on the compact mode are compared with those of that based on the loose one. Also, outcomes of the random colony HDEAs based on each connection mode are compared.

In our experiment, each HDEA should run thirty times independently for each instance under five equal difference migration interval values, respectively. Thus, the total running times should be $4 \times 9 \times 5 \times 30 = 5,400$ ones. However, for lin318 and p654, all the algorithms run under only one interval since they can all obtain the optimal solution at 100 %. The results are shown in Table 3.

According to the t-tests results of Ring colony HDEAs, it can be seen that the algorithm based on compact mode yields significantly better results than the loose connection one in sixteen out of thirty-seven cases, while it only statistically loses in three cases. There are no significant differences in the rest of cases. In the comparison of Random colony HDEAs, the former statistically wins in twelve cases and loses in six ones. These comparisons show that the compact mode is a better choice for HEDAs with our scheme than the other one.

Therefore, in all the following experiments, colony HDEAs are all based on the compact connection mode.

4.5 Experiment to compare general results

The outcomes of the all HDEAs are compared in this experiment. In our plan, these algorithms should be executed thirty times independently for the nine instances under five migration intervals, respectively. Then, the total running times should be as same as those of the connection mode experiment. Nevertheless, for the same reason, lin318 and p654 are solved under the only one migration interval. The results are shown in Table 4.

Based on the results in Table 4, the difficulty of each instance for one HDEA, D, can be expressed according to the following measure:

$$D = \frac{F_{ave} - F_o}{F_o} \tag{4}$$

Table 2 Experiment results of ring DEA and ring colony HDEA

Instance	Migration interval	Sixteen subpopulation (4 × 4)				Thirty-six subpopulation (6 × 6)				Sixty-four subpopulation (8 × 8)			
		Ring individual HDEA		Ring DEA		Ring individual HDEA		Ring DEA		Ring individual HDEA		Ring DEA	
		Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation
lin318	30,000	42,031.8	7.26	42,031.8	7.26	42,029.7	3.83	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00
	40,000	42,031.8	7.26	42,030.4	5.33	42,029.7	3.83	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00
	50,000	42,031.1	6.41	42,033.2	12.66	42,029.7	3.83	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00
	60,000	42,029.7	3.83	42,029.7	3.83	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00
	70,000	42,029.7	3.83	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00	42,029.0	0.00
pcb442	100,000	50,934.6	17.83	50,938.6	7.10	*50,921.7*	*27.38*	*50,936.4*	*8.06*	*50,906.6*	*38.60*	*50,932.6*	*9.26*
	150,000	50,937.0	7.77	50,939.7	6.22	*50,918.4*	*33.40*	*50,934.7*	*8.53*	*50,906.7*	*42.51*	*50,932.4*	*8.51*
	200,000	50,934.3	9.00	50,937.8	7.73	*50,919.0*	*32.80*	*50,932.1*	*10.06*	*50,921.2*	*21.84*	*50,930.3*	*8.76*
	250,000	*50,930.9*	*19.30*	*50,939.4*	*6.60*	50,913.0	43.24	50,927.4	21.36	*50,916.0*	*30.31*	*50,931.3*	*8.28*
	300,000	50,932.6	22.06	50,937.6	8.36	50,908.7	43.01	50,927.0	26.24	50,910.2	28.91	50,918.9	31.45
u574	60,000	*36,948.1*	*24.13*	*36,986.1*	*40.44*	*36,927.8*	*21.37*	*36,950.2*	*24.52*	*36,912.9*	*16.04*	*36,923.7*	*21.17*
	70,000	*36,955.7*	*25.18*	*36,976.0*	*33.02*	*36,930.7*	*24.65*	*36,951.7*	*31.05*	*36,909.6*	*12.25*	*36,930.4*	*21.16*
	80,000	*36,949.8*	*24.64*	*36,977.2*	*42.30*	*36,926.0*	*22.39*	*36,948.0*	*20.96*	*36,911.6*	*15.20*	*36,925.2*	*25.37*
	90,000	*36,941.4*	*26.47*	*36,975.9*	*30.37*	*36,919.7*	*20.68*	*36,948.2*	*19.61*	*36,910.7*	*13.13*	*36,925.7*	*21.06*
	100,000	*36,948.0*	*19.72*	*36,984.6*	*34.35*	*36,922.1*	*22.54*	*36,941.8*	*24.42*	*36,910.6*	*13.02*	*36,919.1*	*18.85*
p654	50,000	34,643.3	1.46	34,643.9	2.08	34,643.0	0.00	34,643.1	0.37	34,643.0	0.00	34,643.0	0.00
	60,000	34,643.1	0.51	34,643.4	1.52	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00
	70,000	34,643.1	0.37	34,643.5	1.81	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00
	80,000	34,643.0	0.00	34,643.2	0.61	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00
	90,000	34,643.0	0.00	34,643.1	0.37	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00	34,643.0	0.00
d657	150,000	*49,075.6*	*30.69*	*49,104.7*	*34.48*	*49,037.5*	*42.50*	*49,076.2*	*33.35*	*49,002.9*	*27.13*	*49,048.6*	*29.36*
	200, 000	*49,067.8*	*32.50*	*49,089.8*	*39.94*	*49,035.4*	*36.51*	*49,062.3*	*41.15*	49,006.9	29.82	49,019.7	32.87
	250,000	*49,062.5*	*40.61*	*49,081.8*	*26.58*	49,022.1	28.50	49,036.2	36.73	*48,991.0*	*30.99*	*49,014.2*	*28.99*
	300,000	49,055.7	35.41	49,071.1	29.40	*49,011.0*	*35.89*	*49,046.6*	*32.24*	*48,976.3*	*31.03*	*49,012.6*	*33.39*
	350,000	49,061.3	44.06	49,071.4	39.20	49,020.4	27.72	49,026.3	35.02	48,996.7	32.30	49,001.0	32.95
u724	150,000	*42,076.8*	*36.00*	*42,121.6*	*41.70*	*42,022.4*	*33.67*	*42,078.9*	*31.69*	*42,011.2*	*33.38*	*42,049.8*	*26.96*
	200,000	*42,074.4*	*32.61*	*42,112.9*	*35.11*	*42,020.7*	*36.47*	*42,055.8*	*30.52*	*42,009.6*	*31.26*	*42,039.1*	*32.37*
	250,000	42,080.8	37.29	42,089.5	37.28	42,032.7	29.25	42,046.3	25.02	*41,990.7*	*23.34*	*42,026.8*	*26.34*
	300,000	*42,067.9*	*36.95*	*42,089.5*	*39.88*	*42,024.9*	*35.29*	*42,045.0*	*32.26*	*41,996.9*	*26.61*	*42,016.6*	*37.29*
	350,000	42,061.8	33.66	42,072.5	42.53	*42,012.6*	*19.66*	*42,032.1*	*33.76*	*41,989.9*	*31.82*	*42,010.5*	*26.81*

Table 2 continued

Instance	Migration interval	Sixteen subpopulation (4 × 4)				Thirty-six subpopulation (6 × 6)				Sixty-four subpopulation (8 × 8)			
		Ring individual HDEA		Ring DEA		Ring individual HDEA		Ring DEA		Ring individual HDEA		Ring DEA	
		Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation
rat783	150,000	8,824.2	8.79	8,824.7	9.40	8,813.9	5.86	8,815.5	8.18	8,807.7	2.51	8,809.4	4.30
	200,000	8,817.9	7.33	8,821.1	8.99	8,812.4	5.98	8,813.3	5.02	8,808.8	3.69	8,807.6	2.50
	250,000	*8,815.8*	*6.59*	*8,819.6*	*6.71*	8,808.6	3.49	8,809.7	4.30	8,807.2	1.47	8,807.6	2.70
	300,000	*8,815.0*	*6.13*	*8,820.0*	*8.64*	8,809.7	3.40	8,809.5	4.37	8,807.1	1.84	8,807.1	1.88
	350,000	8,816.3	7.56	8,816.5	7.45	8,808.8	2.98	8,810.2	4.33	8,806.8	1.00	8,806.9	1.78
dsj1000	600,000	*18,752,623.1*	*19,430.17*	*18,779,587.9*	*24,942.24*	*18,734,405.2*	*13,254.09*	*18,743,877.4*	*16,179.60*	18,718,287.8	15,532.32	18,725,217.3	16,875.44
	800,000	18,753,710.9	30,623.05	18,766,681.6	28,256.50	*18,726,333.3*	*16,657.93*	*18,735,640.1*	*14,653.75*	*18,710,508.2*	*20,606.26*	*18,724,599.2*	*13,152.40*
	1,000,000	*18,744,322.9*	*24,195.68*	*18,763,055.4*	*27,411.22*	18,722,614.8	18,519.60	18,729,860.5	15,437.37	18,714,437.1	17,582.44	18,715,745.1	17,232.00
	1,200,000	18,748,920.8	27,722.07	18,758,048.7	23,265.94	18,727,362.7	13,152.69	18,728,654.0	16,537.92	18,711,657.1	18,901.92	18,712,496.0	20,398.39
	1, 400,000	18,736,646.7	23,242.55	18,757,767.8	26,774.19	*18,717,367.1*	*18,823.32*	*18,729,028.6*	*20,676.25*	*18,700,377.4*	*21,684.72*	*18,716,388.7*	*21,529.82*
pr1002	500,000	259,587.4	282.27	259,734.9	293.82	259,247.7	117.34	259,283.4	160.35	*259,155.4*	*40.87*	*259,191.9*	*71.03*
	600,000	*259,472.8*	*254.46*	*259,613.0*	*262.65*	259,214.4	105.45	259,274.4	145.66	259,157.1	65.55	259,182.9	79.39
	700,000	259,529.2	259.69	259,545.4	190.58	259,209.9	119.99	259,236.0	117.50	259,150.3	49.36	259,171.0	30.09
	800,000	259,441.4	186.03	259,472.0	203.85	259,236.5	123.76	259,245.3	106.46	259,145.2	36.58	259,161.4	28.31
	900,000	259,403.0	188.95	259,488.7	259.54	259,178.8	74.67	259,207.9	81.85	259,141.1	30.81	259,149.3	53.29

All cases which have significant difference in terms of the t-test with 95 % confidence are highlighted by italics

Table 3 Experiment result with different connection modes

Instance	Migration interval	Ring colony HDEAs				Random colony HDEAs			
		Compact connection		Loose connection		Compact connection		Loose connection	
		Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation
lin318	50,000	42,029	0	42,029	0	42,029	0	42,029	0
pcb442	100,000	*50,882.6*	*48.62*	*50,920.9*	*26.93*	*50,906.5*	*32.15*	*50,927.2*	*7.54*
	150,000	*50,896.5*	*50.26*	*50,919.4*	*27.72*	*50,905.7*	*37.33*	*50,922.9*	*19.64*
	200,000	50,895.6	52.24	50,914	32.34	50,906.4	42.88	50,916.3	29.33
	250,000	*50,890.3*	*57.38*	*50,915.3*	*22.67*	*50,909.7*	*39.01*	*50,924.8*	*8.51*
	300,000	50,901.4	45.9	50,901.8	48.86	50,905.8	41.63	50,920.6	31.93
u574	60,000	36,921	22.04	36,915.2	17.46	36,945.5	35.91	36,929.4	26.56
	70,000	36,923.4	19.22	36,916.4	20.15	36,935.7	29.42	36,927	24.89
	80,000	36,914.1	18.52	36,912.6	16.01	36,934.6	21.66	36,928.7	23.55
	90,000	*36,920.3*	*20.78*	*36,910.4*	*12.43*	36,929.1	24.23	36,928.1	22.68
	100,000	36,912.9	16.25	36,908.5	10.96	36,925.1	22.4	36,925.9	24.16
p654	80,000	34,643	0	34,643	0	34,643	0	34,643	0
d657	150,000	*48,976.5*	*25.99*	*49,002.7*	*32.52*	49,015.3	34.58	49,009.1	34.75
	200,000	*48,976.5*	*27.44*	*49,005.1*	*31.62*	*48,989.3*	*28.29*	*49,014.9*	*32.95*
	250,000	48,975.6	23.52	48,987.9	24.6	48,986.5	36.1	48,998.5	26.62
	300,000	48,977.3	29.91	48,987.9	33.81	48,990.4	29.13	48,989.1	34.46
	350,000	48,985.3	29.15	48,998	22.29	*48,968.5*	*23.16*	*48,982.5*	*27.07*
u724	150,000	*41,981.1*	*30.84*	*42,023.8*	*27.36*	*42,003.6*	*37.56*	*42,046.6*	*33.86*
	200,000	*41,978.9*	*31.91*	*42,019.4*	*23.88*	*41,994.7*	*28.13*	*42,031.8*	*32.95*
	250,000	*41,970.6*	*24.16*	*41,998*	*22.88*	*41,986*	*29.5*	*42,010.4*	*33.04*
	300,000	*41,981.1*	*28.8*	*41,999.3*	*23.82*	*41,970.3*	*22.76*	*41,997.4*	*34.84*
	350,000	*41,977.2*	*25.65*	*42,005.5*	*20.62*	41,982.1	23.9	41,986.9	24.89
rat783	150,000	*8,815.9*	*7.95*	*8,807.9*	*2.91*	*8,825.9*	*14.4*	*8,811*	*3.95*
	200,000	*8,810.2*	*4.46*	*8,807.6*	*1.98*	*8,815.3*	*9.21*	*8,807.7*	*2.33*
	250,000	8,808.1	3.16	8,807.1	2.47	*8,809.3*	*3.29*	*8,807.8*	*2.62*
	300,000	8,807.5	3.23	8,806.5	0.57	*8,808.2*	*4.68*	*8,806.4*	*0.73*
	350,000	8,807.1	2.03	8,806.8	1.1	*8,807.6*	*2.75*	*8,806.5*	*0.57*
dsj1000	600,000	*18,689,392.7*	*16,934.91*	*18,719,209.7*	*20,396.27*	*18,689,889.4*	*19,173.53*	*18,724,323.5*	*18,985.81*
	800,000	*18,689,379.9*	*13,988.01*	*18,714,875.9*	*16,606.94*	*18,700,320.8*	*21,743.09*	*18,721,241.2*	*17,283.64*
	1,000,000	*18,688,818.8*	*18,237.78*	*18,714,415.5*	*18,981.57*	*18,693,756.6*	*20,380.07*	*18,712,631.9*	*22,271.71*
	1,200,000	*18,687,211.6*	*19,654.14*	*18,716,078*	*17,034.08*	18,694,706.1	20,104.26	18,705,307.8	22,037.64
	1,400,000	*18,700,730.2*	*22,360.31*	*18,714,315.1*	*20,845.02*	18,695,557.9	21,163.01	18,699,476.9	22,103.78
pr1002	500,000	259,149	76.1	259,164.7	63.81	*259,220.2*	*146.31*	*259,158.8*	*42.98*
	600,000	259,141	74.19	259,145	40.07	259,146.9	56.8	259,144.3	32.2
	700,000	259,133.6	65.21	259,144.6	34.49	259,145.5	62.66	259,146.8	34.21
	800,000	*259,131.4*	*37.5*	*259,151.3*	*29.39*	259,131.6	38.01	259,135.3	27.11
	900,000	259,129.7	34.91	259,140.9	28.42	259,138.6	46.17	259,139.5	29.88

All cases which have significant difference in terms of the t-test with 95 % confidence are highlighted by italics

where F_{ave} is the average of all solutions of a HDEA for one instance, F_o is the optimal solution provided by Reinelt (1996). Then, the difficulties are given in Table 5. $\overline{D}$ in the table denotes the average difficulties of each instance for all the algorithms.

According to Table 4, outcomes of lin318 and p654 can be excluded from comparison because, for the all algorithms, the average of solutions for the two instances equals to their optimal solution. It is should be noted in the table that the solutions' standard deviation of the ring colony HDEA is

Table 4 Comparison on general result of different algorithms

Instance/ optimal solution	Migration interval	Ring colony HDEA (I)		Random colony HDEA (II)		Ring individual HDEA (III)		Random individual HDEA (IV)		Solutions have significant difference				
		Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	I–II	I–III	I–IV	II–III	II–IV
lin318/42029	50,000	42,029.0[a]	0	42,029.0[a]	0	42,029.0[a]	0	42,029.0[a]	0	No	No	No	No	No
pcb442/50778	100,000	50,882.6[a]	48.62	50,906.5	32.15	50,906.6[a]	38.6	50,917.2	30.89	*Yes*	*Yes*	*Yes*	No	No
	150,000	50,896.5[a]	50.26	50,905.7[a]	37.33	50,906.7[a]	42.51	50,910.9	32.31	No	No	No	No	No
	200,000	50,895.6[a]	52.24	50,906.4[a]	42.88	50,921.2	21.84	50,913.7	31.9	No	*Yes*	No	No	No
	250,000	50,890.3[a]	57.38	50,909.7[a]	39.01	50,916	30.31	50,912.0[a]	35.4	No	*Yes*	No	No	No
	300,000	50,901.4	45.9	50,905.8[a]	41.63	50,910.2	28.91	50,899.8[a]	50.8	No	No	No	No	No
u574/36910	60,000	36,921.0[a]	22.04	36,945.5[a]	35.91	36,912.9[a]	16.04	36,927.8[a]	23.42	*Yes*	No	No	*Yes*	*Yes*
	70,000	36,923.4[a]	19.22	36,935.7[a]	29.42	36,909.6[a]	12.25	36,916.6[a]	18.52	No	*Yes*	No	*Yes*	*Yes*
	80,000	36,914.1[a]	18.52	36,934.6[a]	21.66	36,911.6[a]	15.2	36,919.5[a]	19.81	*Yes*	No	No	*Yes*	*Yes*
	90,000	36,920.3[a]	20.78	36,929.1[a]	24.23	36,910.7[a]	13.13	36,918.9[a]	22.13	No	*Yes*	No	*Yes*	No
	100,000	36,912.9[a]	16.25	36,925.1[a]	22.4	36,910.6[a]	13.02	36,910.6[a]	13.02	*Yes*	No	No	*Yes*	*Yes*
p654/34643	80,000	34,643.0[a]	0	34,643.0[a]	0	34,643.0[a]	0	34,643.0[a]	0	No	No	No	No	No
d657/48912	150,000	48,976.5	25.99	49,015.3	34.58	49,002.9	27.13	48,994.1	32.74	*Yes*	*Yes*	*Yes*	No	*Yes*
	200,000	48,976.5	27.44	48,989.3	28.29	49,006.9	29.82	49,005.4	33.39	No	*Yes*	*Yes*	*Yes*	*Yes*
	250,000	48,975.6	23.52	48,986.5	36.1	48,991	30.99	48,995.8	30.94	No	*Yes*	*Yes*	No	No
	300,000	48,977.3	29.91	48,990.4	29.13	48,976.3	31.03	48,996.3	24.82	No	No	*Yes*	No	No
	350,000	48,985.3	29.15	48,968.5	23.16	48,996.7	32.3	48,988	27.3	*Yes*	No	No	*Yes*	*Yes*
u724/41910	150,000	41,981.1	30.84	42,003.6	37.56	42,011.2	33.38	42,006.1	29.61	*Yes*	*Yes*	*Yes*	No	No
	200,000	41,978.9	31.91	41,994.7	28.13	42,009.6	31.26	42,012.6	30.82	*Yes*	*Yes*	*Yes*	No	*Yes*
	250,000	41,970.6	24.16	41,986	29.5	41,990.7	23.34	41,996.8	25.56	*Yes*	*Yes*	*Yes*	No	No
	300,000	41,981.1	28.8	41,970.3	22.76	41,996.9	26.61	41,996	27.02	No	*Yes*	*Yes*	*Yes*	*Yes*
	350,000	41,977.2	25.65	41,982.1	23.9	41,989.9	31.82	41,993.4	30.35	No	No	*Yes*	No	No
rat783/8806	150,000	8,815.9	7.95	8,825.9	14.4	8,807.7[a]	2.51	8,808.4[a]	3.23	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
	200,000	8,810.2[a]	4.46	8,815.3[a]	9.21	8,808.8[a]	3.69	8,819.4	15.49	*Yes*	No	*Yes*	*Yes*	No
	250,000	8,808.1[a]	3.16	8,809.3[a]	3.29	8,807.2[a]	1.47	8,807.5[a]	2.37	No	No	No	*Yes*	*Yes*
	300,000	8,807.5[a]	3.23	8,808.2[a]	4.68	8,807.1[a]	1.84	8,807.4[a]	1.76	No	No	No	No	No
	350,000	8,807.1[a]	2.03	8,807.6[a]	2.75	8,806.8[a]	1	8,806.9[a]	1.53	No	No	No	No	No
dsj1000/ 18659688	600,000	18,689,392.7	16,934.9	18,689,889.4	19,173.53	18,718,287.8	15,532.3	18,720,375.5	18,985.41	No	*Yes*	*Yes*	*Yes*	*Yes*
	800,000	18,689,379.9	13,988	18,700,320.8	21,743.09	18,710,508.2	20,606.3	18,720,595.1	16,852.17	*Yes*	*Yes*	*Yes*	No	*Yes*
	1,000,000	18,688,818.8	18,237.8	18,693,756.6	20,380.07	18,714,437.1	17,582.4	18,711,642.7	19,564.78	No	*Yes*	*Yes*	*Yes*	*Yes*
	1,200,000	18,687,211.6	19,654.1	18,694,706.1	20,104.26	18,711,657.1	18,901.9	18,708,625.5	22,595.97	No	*Yes*	*Yes*	*Yes*	*Yes*

Table 4 continued

Instance/optimal solution	Migration interval	Ring colony HDEA (I)		Random colony HDEA (II)		Ring individual HDEA (III)		Random individual HDEA (IV)		Solutions have significant difference				
		Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	Average	Standard deviation	I–II	I–III	I–IV	II–III	II–IV
pr1002/259045	500,000	259,149.0	76.1	259,220.2[a]	146.31	259,155.4	40.87	259,180.9	83.26	*Yes*	No	No	*Yes*	No
	600,000	259,141.0[a]	74.19	259,146.9[a]	56.8	259,157.1[a]	65.55	259,140.1[a]	35.25	No	No	No	No	No
	700,000	259,133.6[a]	65.21	259,145.5	62.66	259,150.3[a]	49.36	259,159.1	34.54	No	No	No	No	No
	800,000	259,131.4[a]	37.5	259,131.6[a]	38.01	259,145.2	36.58	259,155.0[a]	47.78	No	No	*Yes*	No	*Yes*
	900,000	259,129.7[a]	34.91	259,138.6[a]	46.17	259,141.1	30.81	259,142.4[a]	39.16	No	No	No	No	No
pr1002/259045	500,000	259,149.0	76.1	259,220.2[a]	146.31	259,155.4	40.87	259,180.9	83.26	*Yes*	No	No	*Yes*	No
	600,000	259,141.0[a]	74.19	259,146.9[a]	56.8	259,157.1[a]	65.55	259,140.1[a]	35.25	No	No	No	No	No
	700,000	259,133.6[a]	65.21	259,145.5	62.66	259,150.3[a]	49.36	259,159.1	34.54	No	No	No	No	No
	800,000	259,131.4[a]	37.5	259,131.6[a]	38.01	259,145.2	36.58	259,155.0[a]	47.78	No	No	*Yes*	No	*Yes*
	900,000	259,129.7[a]	34.91	259,138.6[a]	46.17	259,141.1	30.81	259,142.4[a]	39.16	No	No	No	No	No

All cases which have significant difference in terms of the t-test with 95 % confidence are highlighted by italics

[a] Denotes that the optimal solution can be obtained in this 30 times running

larger than that of the ring individual one in 27 cases out of 35 ones and than that of the random individual one in 18 ones. Similarly, that of random colony HDEA is larger than that of the two traditional algorithms in 25 cases and in 26 ones, respectively. The results show that the stability of the colony HDEAs is not better than the traditional algorithms.

It can be seen from Table 5 that the size of a TSP instance is not the decisive factor of its difficulty for a HDEA. Furthermore, according to the value of $\overline{D}$, dsj1000, u724, pcb442 and d657 are four high difficulty instances while others belong to low difficulty ones.

In Table 4, for high difficulty instances, the ring colony HDEA yields significantly better outcomes than the ring individual one in 14 cases out of 20 cases and than the random individual one in 13 ones. Meanwhile, there are no significant differences in the rest of cases. However, for the three low difficulty instances, rat783, u574 and pr1002, the ring colony HDEA never statistically wins the traditional algorithms. Moreover, it even statistically loses the former of the traditional algorithms in three cases and the latter in one case. On the other hand, the random colony HDEA products significantly better outcomes for high difficulty instances than the two traditional algorithms in six cases and eight ones, respectively. Meanwhile, it obtains significantly worse ones than the random individual HDEA in a case. Nonetheless, for instances with low difficulty, it statistically wins the ring individual HDEA in only one case but statistically loses this traditional algorithm in nine cases and the other one in six cases. It also can be seen in Table 4 that the outcomes of the ring colony HDEA are significantly better than those of the random colony one in 11 cases out of 35 ones and significantly worse in only one case.

For high difficulty instances, the results show the superiority of colony HDEAs. The reason why the both colony HDEAs are hard to obtain significantly better outcomes for low difficulty instances than the individual ones is as follow. According to Table 4, the colony HDEAs have no advantage in stability. As Table 4 shows, all the algorithms can get the optimal solution of the low difficulty instances. Then, for these instances, stability becomes a more important factor to decide the quality of solutions. Hence, the colony HDEAs statistically lose in many cases of low difficulty instances. In addition, the random colony HDEA is outperformed by the ring colony one because the slightly larger global migration size is still not enough to compensate conflicting migration arising from the narrow range of choices on migration object in random global migration with moving colony.

5 Concluding remarks

In this paper, we have presented the global migration strategy with moving colony for HDEAs. In global migration based on

Table 5 Experiment result of difficulty for HDEAs

	lin318	pcb442	u574	p654	d657	u724	Rat783	dsj1000	pr1002
F_{ave} of I	42,029.0	50,893.3	36,918.3	34,643.0	48,978.2	41,977.8	8,809.7	18,691,106.6	259,136.9
F_{ave} of II	42,029.0	50,906.8	36,934.0	34,643.0	48,990.0	41,987.3	8,813.3	18,694,846.2	259,156.6
F_{ave} of III	42,029.0	50,912.2	36,911.1	34,643.0	48,994.8	41,999.7	8,807.5	18,711,053.5	259,149.8
F_{ave} of IV	42,029.0	50,910.7	36,918.7	34,643.0	48,995.9	42,001.0	8,809.9	18,713,973.7	259,155.5
F_o	42,029	50,778	36,905	34,643	48,912	41,910	8,806	18,659,688	259,045
D_{I}	0.00000	0.00227	0.00036	0.00000	0.00135	0.00162	0.00043	0.00168	0.00035
D_{II}	0.00000	0.00254	0.00079	0.00000	0.00160	0.00185	0.00082	0.00188	0.00043
D_{III}	0.00000	0.00264	0.00017	0.00000	0.00169	0.00214	0.00017	0.00275	0.00040
D_{IV}	0.00000	0.00261	0.00037	0.00000	0.00172	0.00217	0.00045	0.00291	0.00043
$\overline{D}$	0.00000	0.00252	0.00042	0.00000	0.00159	0.00194	0.00047	0.00231	0.00040

the proposed strategy, subpopulations move between groups. Such global migration can be realized without any communication cost. Based on different global migration topology, two kinds of global migration based on the proposed strategy are employed in HDEAs.

In our experiment, nine benchmark instances of the TSP are used to test algorithms. Firstly, the outcomes of the ring DEA and those of the ring individual HDEA are compared under three different values of the subpopulation quantity. The results show that the algorithm with hierarchy has a significant advantage on solutions. Then, comparison on outcomes of colony HDEAs based on different connection modes shows that the compact mode is the better choice for HDEAs with our scheme. After that, the HDEAs based on the purposed scheme and the traditional ones are compared. The results show that the ring colony HDEA obtains significantly better solutions than traditional ones on the high difficulty instances while the random one have not so great an advantage.

According to the facts listed above, some conclusions can be given as below:

- Solutions of both HDEAs and simple DEAs will be better and better with the increase of the subpopulation quantity.
- A HDEA based on the same EA having the same quantity of subpopulations can be regarded as an useful upgrade of a DEA without any extra cost.
- Random topology is not reasonable for global migration with moving colony for conflicting migration caused by the narrow range of choices on migration object. Though this problem may be solved by further increasing the global migration size, more computing resources will be wasted.
- If a global topology without randomness, such as ring, is used, global migration with moving colony is a significantly better choice for HDEAs than the traditional method when a problem is difficult enough. In addition, this scheme needs no communication between master process and computing processes.

Though realizing multi-level HDEAs is feasible (Herrera et al. 1999), it is hard to find any implementation in literature. In such a HDEA, higher level migration should occur at a lower frequency. If it is based on the individual migration strategy, the function of higher level migration is more limited for its fewer running times. However, in a multi-level HDEA based on the global migration strategy with moving colony, migration object is a colony whose level is one grade lower than the level of colonies between which migration is executed. Higher level migration does occur at a lower frequency, but is greater in grain size of its migration object. Such HDEAs deserve further investigation. This issue should be addressed in the future.

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References

Alba E, Tomassini M (2002) Parallelism and evolutionary algorithms. IEEE Trans Evol Comput 6(5):443–462

Arnaldo I, Contreras I, Millán-Ruiz D, Hidalgo JI, Krasnogor N (2013) Matching island topologies to problem structure in parallel evolutionary algorithms. Soft Comput 17(7):1209–1225

Cai ZH, Peng JG, Gao W, Wei W, Kang LS (2005) An improved evolutionary algorithm for the traveling salesman problem. Chin J Comput 28(5):823–828

Cantu-Paz E (1998) A survey of parallel genetic algorithms. Calculateurs Parallles, Reseaux et Systems Repartis 10(2):141–171

Choi JN, Oh SK, Pedrycz W (2009) Identification of fuzzy relation models using hierarchical fair competition-based parallel genetic algorithms and information granulation. Appl Math Model 33(6):2791–2807

Cohoon JP, Hegde SU, Martin WN, Richards D (1987) Punctuated equilibria: a parallel genetic algorithm. In: Proceedings of the second international conference on genetic algorithms on genetic algorithms and their application, pp 148–154

Fernandez F, Tomassini M, Vanneschi L (2001) Studying the influence of communication topology and migration on distributed genetic programming. In: Proceedings of the annual conferences on genetic programming, pp 51–63

Goldberg DE, Lingle R (1985) Alleles, loci and the TSP. In: Proceedings of first international conference on genetic algorithms, pp 154–159
Gonzalez LF, Lee DS, Srinivas K, Wong KC (2006) Single and multi-objective UAV aerofoil optimisation via hierarchical asynchronous parallel evolutionary algorithm. Aeronaut J 110(1112):659–672
Herrera F, Lozano M, Moraga C (1999) Hierarchical distributed genetic algorithms. Int J Intell Syst 14(11):1099–1121
Karakasis MK, Koubogiannis DG, Giannakoglou KC (2007) Hierarchical distributed metamodel-assisted evolutionary algorithms in shape optimization. Int J Numer Methods Fluids 53(3):455–469
Lässig J, Sudholt D (2013) Design and analysis of migration in parallel evolutionary algorithms. Soft Comput 17(7):1121–1144
Lee DS, Periaux J, Onate E, Gonzalez LF, Qin N (2011) Active transonic aerofoil design optimization using robust multiobjective evolutionary algorithms. J Aircr 48(3):1084–1094
Liakopoulos PIK, Kampolis IC, Giannakoglou KC (2008) Grid enabled, hierarchical distributed metamodel-assisted evolutionary algorithms for aerodynamic shape optimization. Future Gener 24(7):701–708
Lin S, Kernighan BW (1973) An effective heuristic algorithm for the traveling-salesman problem. Oper Res 21:498–516
Power D, Ryan C, Azad RMA (2005) Promoting diversity using migration strategies in distributed genetic algorithms. In: Proceedings of IEEE congress on evolutionary computation, vol 2, pp 1831–1838
Reinelt G (1996) TSPLIB. University of Heidelberg, Heidelberg
Skolicki Z, De Jong K (2005) The influence of migration sizes and intervals on island models. In: Proceedings of genetic and evolutionary computation conference, pp 1295–1302
Tao G, Michalewicz Z (1998) Evolutionary algorithms for the TSP. In: Proceedings of the 5th parallel problem solving from nature conference, pp 803–812
Yuan XL, Bai Y (2010) Identifying stochastic nonlinear dynamic systems using multi-objective hierarchical fair competition parallel genetic programming. J Mult Valued Log Soft Comput 16(6):643–660